BLUESTOCK FINTECH

Mutual Fund Analytics Platform - Final Project Report
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Technology Stack:

- Python, Pandas, NumPy
- PostgreSQL, SQLAlchemy
- Power BI, Jupyter Notebook

1. Executive Summary

The Mutual Fund Analytics Platform was developed to provide a comprehensive analytics solution for evaluating Indian mutual funds using historical and transactional data. The project integrates multiple datasets, including NAV history, investor transactions, portfolio holdings, benchmark indices, SIP inflows, AUM history, and fund performance, into a unified analytics platform.

The project covers the complete data analytics lifecycle, including data extraction, cleaning, transformation, loading into PostgreSQL, exploratory data analysis, financial performance evaluation, advanced risk analytics, and interactive dashboard development using Power BI. The platform enables investors and analysts to compare funds using various performance and risk metrics and supports data-driven investment decisions.

2. Data Overview

The project uses multiple datasets representing different aspects of the mutual fund industry.

Data Sources

- Fund Master
- NAV History
- Scheme Performance

- Investor Transactions
- Monthly SIP Inflows
- Category Inflows
- Portfolio Holdings
- Benchmark Indices
- AUM History
- Folio History

Total Datasets

- 10 CSV files
- Live NAV data fetched using MFAPI
- PostgreSQL Database
- Power BI Dashboard

Key Features

- Historical NAV data
- Investor transaction records
- Portfolio allocation
- Benchmark performance
- SIP inflows
- Industry AUM
- Folio growth

3. ETL Process

The project follows a complete ETL (Extract, Transform, Load) pipeline.

Extract

- Imported multiple CSV datasets using Pandas.
- Retrieved live NAV data using MFAPI.
- Loaded benchmark and industry datasets.

Transform

Several data cleaning operations were performed:

- Removed duplicate records.
- Converted date columns into datetime format.
- Standardized transaction types.
- Forward-filled missing NAV values.
- Validated expense ratios.
- Calculated daily returns.
- Generated cleaned datasets for analysis.

Load

The cleaned datasets were loaded into PostgreSQL using SQLAlchemy.

Relationships were created using:

- `amfi_code`
- `date`

The database was designed using a Star Schema with dimension and fact tables for efficient querying.

4. EDA Findings

Exploratory Data Analysis (EDA) was performed to understand fund performance, investor behaviour, and industry trends.

Visualizations Created

- Industry AUM Growth (2022–2025)
- Monthly SIP Trend
- Category Inflow Heatmap
- Investor Demographics
- Geographic Distribution
- State-wise Investment
- Age Group Distribution
- Gender Distribution
- Portfolio Sector Allocation
- NAV Correlation Matrix
- Benchmark Trends
- Folio Growth

- Risk vs Return Scatter Plot
- Fund Scorecard
- Benchmark Comparison

More than **15 charts** were developed and exported as PNG images for reporting.

Key Insights

- Industry AUM increased consistently between 2022 and 2025.
- Monthly SIP inflows reached record levels in late 2025.
- Equity-oriented categories attracted the highest inflows.
- Adult and middle-aged investors represented the largest investor segment.
- Tier-1 cities contributed the highest investment volume.
- Portfolio diversification varied significantly across equity funds.
- Risk and return showed a positive relationship.
- Investor participation increased steadily over the study period.

5. Performance Analysis

Advanced financial analytics were performed for all mutual funds.

Metrics Calculated

Return Metrics

- Daily Return
- CAGR (1 Year)
- CAGR (3 Year)
- CAGR (5 Year)

Risk Metrics

- Standard Deviation
- Sharpe Ratio
- Sortino Ratio
- Maximum Drawdown
- Historical VaR (95%)
- Conditional VaR (CVaR)
- Rolling 90-Day Sharpe Ratio

Market Metrics

- Alpha
- Beta
- Tracking Error
- Benchmark Comparison

Investor Analytics

- Investor Cohort Analysis
- SIP Continuity Analysis
- Fund Recommendation System
- Sector HHI Concentration

These metrics provide a comprehensive evaluation of mutual fund performance beyond simple returns.

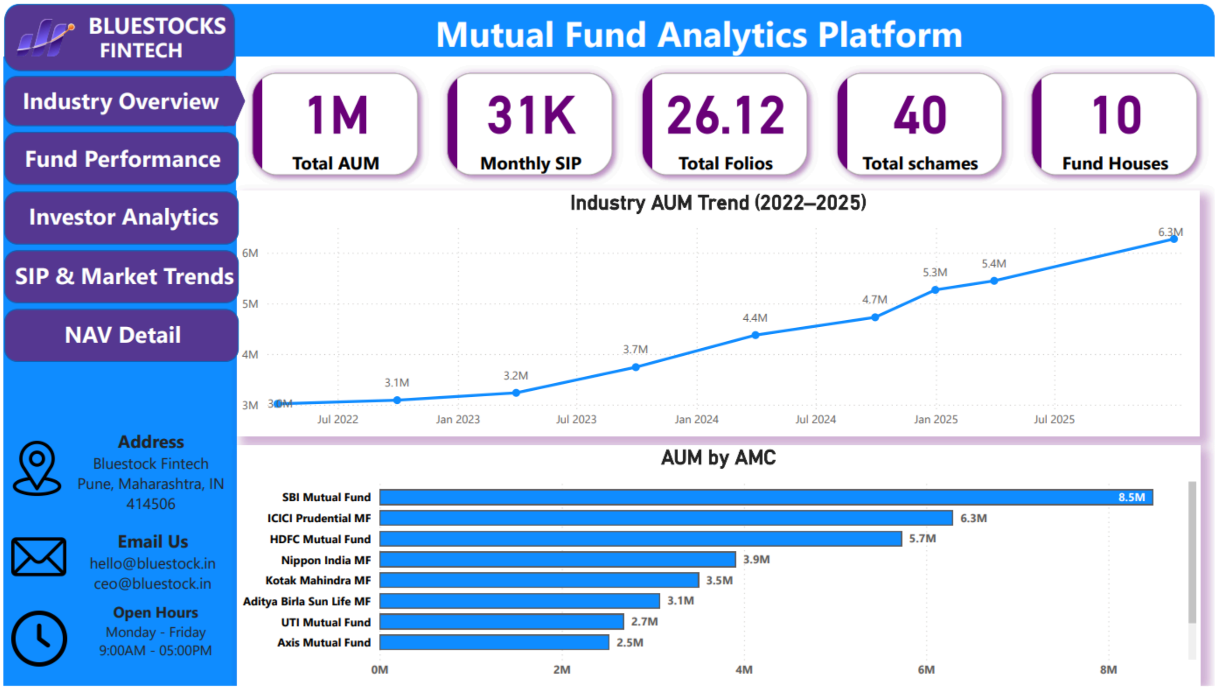
6. Dashboard Screenshots

The project includes a four-page interactive Power BI dashboard.

Page 1 – Industry Overview

Key Elements Included:

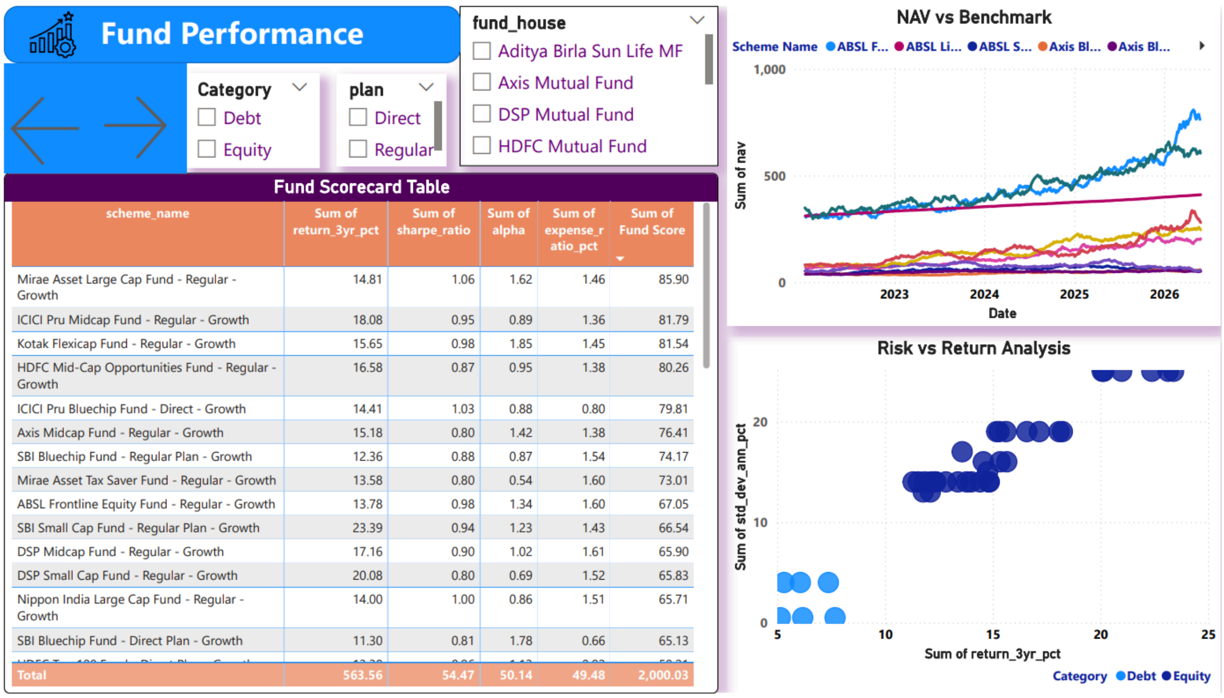
- Total AUM
- Monthly SIP
- Total Folios
- Total Schemes
- Industry AUM Trend
- AUM by AMC



Page 2 – Fund Performance

Key Elements Included:

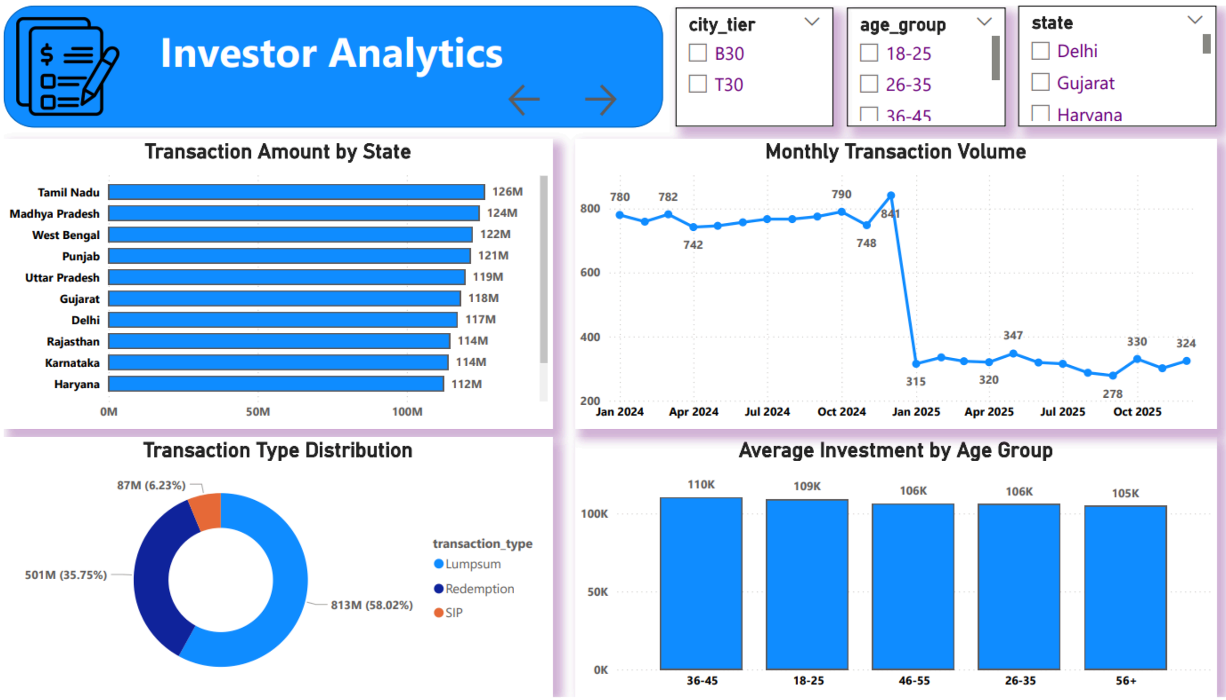
- Risk vs Return Scatter Plot
- NAV vs Benchmark
- Fund Scorecard
- Category Filters
- Plan Filters
- Fund House Filters



Page 3 – Investor Analytics

Key Elements Included:

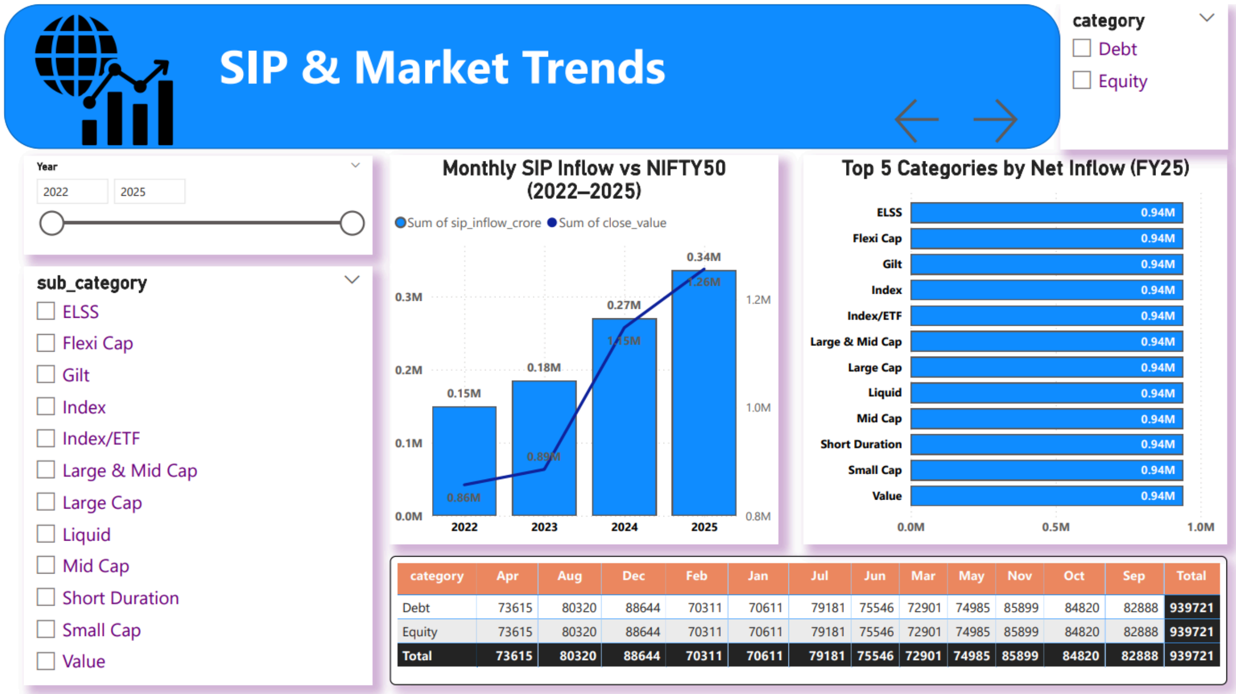
- Transaction Amount by State
- SIP/Lumpsum/Redemption Distribution
- Age Group Analysis
- Monthly Transaction Trend
- City Tier, Age Group and State Filter



Page 4 – SIP & Market Trends

Key Elements Included:

- SIP vs Nifty Trend
- Category Inflow Heatmap
- Top Categories
- Market Trends



7. Recommendations

Based on the analysis, the following recommendations are suggested:

1. Promote SIP investments among younger investors to improve long-term participation.
2. Identify and engage investors with irregular SIP contributions to improve retention.
3. Encourage portfolio diversification for highly concentrated funds.
4. Use risk-adjusted metrics, such as the Sharpe Ratio and Alpha, when selecting mutual funds.
5. Provide personalised fund recommendations based on investor risk appetite.

8. Limitations

The project has the following limitations:

1. Analysis depends on historical data and may not predict future market performance.
2. Some assumptions were made while handling missing values during data cleaning.

3. Benchmark data availability affects Alpha and Beta estimation.
4. Portfolio holdings represent snapshots and may change over time.
5. The dashboard uses periodic data refresh instead of real-time streaming.

9. Conclusion

The Mutual Fund Analytics Platform successfully demonstrates a complete end-to-end data analytics workflow. The project integrates ETL processes, database design, exploratory analysis, financial performance evaluation, advanced risk analytics, and interactive dashboard visualization into a single platform.

By combining multiple financial metrics with investor behaviour analysis, the platform provides meaningful insights that support informed investment decisions. The developed dashboard and analytical models can assist investors, analysts, and financial institutions in evaluating mutual funds more effectively and identifying investment opportunities based on both return and risk characteristics.